

WANSHIKA PATRO

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SUMMARY

Data Analyst with 5 years of experience supporting data-driven decision-making across Indian and US organizations. Skilled in transforming raw datasets into actionable insights, building automated reporting assets, and creating scalable analytical workflows. Strong background in BI development, SQL engineering, predictive modeling, and cloud-based warehouse environments supporting business operations and strategic initiatives.

SKILLS

Programming & Querying: Python (Pandas, NumPy, Matplotlib, Seaborn, Streamlit), R, SQL, DAX, M Language

Data Processing: ETL Development, Data Normalization, Transformation Logic, Power Query, Batch and Incremental Refresh

Visualization & BI Tools: Power BI, Tableau, Excel (Advanced Functions, VBA Automation), Interactive Dashboarding, Trend Analytics

Cloud & Storage Systems: Snowflake, BigQuery, AWS Redshift, Azure, Dimensional Modeling, Warehouse Optimization

Database Platforms: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB

Statistical & ML Techniques: Regression Models, Hypothesis Tests, Segment Analysis, Experimental Design, Forecast Pattern Recognition

Utilities & Integrations: Git, JIRA, REST-based Data Pull, API Parsing, Structured/Unstructured File Handling

EXPERIENCE

Northern Trust, United States

January 2025– Present

Senior Data Analyst

- Developed SQL-driven analytical models and aggregation layers to enhance performance dashboards, reducing data processing time by 35% for business stakeholders.
- Built ETL refresh sequences using Power Query and Python, consolidating data from CRM, ticketing systems, and marketing platforms into unified reporting structures.
- Designed Power BI visualizations with custom DAX logic, interactive filters, and KPI summaries enabling managers to track productivity, revenue flow, and operational exceptions.
- Conducted systematic data validation and profiling exercises, improving accuracy and consistency across 90% of legacy datasets.
- Implemented statistical evaluations including trend breakdowns, seasonality detection, and distribution checks improving forecasting reliability for planning teams.
- Created dimensional data models and optimized schema layouts to support multi-layered BI queries with enhanced refresh stability.
- Delivered ad-hoc research reports and segmentation analysis helping leadership understand behavioral patterns and operational bottlenecks.

Space Infolab, India

January 2020– December 2023

Data Analyst

- Constructed scalable data pipelines integrating Snowflake and BigQuery environments, implementing query optimization and load strategies for large-volume datasets.
- Produced Tableau scorecards for operations, product insights, and executive monitoring enabling rapid visibility into performance metrics and real-time health indicators.
- Automated data processing routines using Python and API feeds reducing manual intervention effort by 55% across weekly reporting tasks.
- Performed experimental evaluations using statistical methods, uplift measurement, and segment-level analysis to support product enhancement initiatives.
- Developed forecasting visuals and forward-looking analytics providing leadership with improved revenue and demand projections.
- Strengthened warehouse governance by defining metric logic, improving dataset lineage visibility, and coordinating changes with cross-functional US teams.
- Implemented structured documentation, version-controlled SQL artifacts, and repeatable quality checks enhancing reliability of recurring analytics cycles.

EDUCATION

Master of Science in Data Science

January 2024– May 2025

University at Buffalo, Buffalo, NY

Bachelor of Computer Science

July 2017– August 2021

Kalinga Institute of Industrial Technology, Bhubaneswar, Odisha

PROJECTS

Revenue Forecasting Dashboard | Python, XGBoost, Tableau, Streamlit

- Built an end-to-end forecasting pipeline over 145K+ transactional records, improving overall prediction accuracy by 18%.
- Developed interactive Tableau and Streamlit dashboards supporting anomaly detection, trend monitoring, and planning workflows for 20+ business stakeholders.
- Delivered forecast-driven insights that optimized inventory distribution processes, yielding a 10% improvement in allocation efficiency.

Crime Trend Analytics and Prediction | Python, SQL, Tableau, Matplotlib

- Processed and engineered features from 1M+ crime incident records, attaining 83% forecasting accuracy across temporal and geographic patterns.
- Designed predictive heatmaps and spatial visualizations that enhanced resource deployment efficiency by 15%.
- Presented analytical findings that contributed to the formulation of two public safety initiatives.

Stock Price Prediction and Market Simulation | Python, Random Forest, SVM, Pandas

- Built predictive models leveraging technical indicators and achieved 82% accuracy in short-term movement forecasting.
- Constructed a simulation framework testing over 10 failure and stress scenarios to evaluate model stability.
- Compared batch versus streaming inference pipelines, implementing optimizations that decreased inference latency by 35%.

AI-Driven Recommendation Engine | Python, Flask, Recommender Models

- Developed a hybrid recommendation model combining collaborative and content-based methods, achieving approximately 25% engagement uplift during controlled testing.
- Deployed real-time recommendation services through a Flask API, maintaining sub-200ms latency under typical load.
- Defined performance KPIs and executed cohort-level experimentation that improved CTR by 15%.

Toxic Comment Detection (LLM/NLP) — Python, Hugging Face

- Implemented an NLP moderation tool integrating BERT-based classification, reducing manual review workload by 30%.
- Improved model precision by 12% over traditional logistic regression baselines through advanced transformer fine-tuning.
- Added 50+ structured feedback loops to refine prediction thresholds and enhance end-user adoption across teams.